PAUL SIMIYU NABENDE

Senior Software Engineer | Backend & Distributed Systems | Architecture | Applied AI
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PROFESSIONAL SUMMARY

Senior Software Engineer with 8+ years of experience architecting and delivering high-throughput backend systems, event-driven microservices, and mobile platforms across fintech, digital payments, and agritech. Recognized for end-to-end domain ownership, spanning systemic architecture, reliability engineering, production incident leadership, and engineering mentorship. Expert in Java, Spring Boot, Kotlin, TypeScript, AWS, Kubernetes, PostgreSQL, and security-first system design. Experienced in applied AI research and rapid prototyping for developer workflows, compliance, and security tooling.

CORE COMPETENCIES

- **Languages:** Java, Kotlin, TypeScript, JavaScript, C#, SQL
- **Backend & Architecture:** Spring Boot, Spring Cloud, Microservices, Event-Driven Architecture, REST APIs, Webhooks, API Gateways, Message Queues, System Design, Technical Design Reviews, Proof-of-Concept (PoC)
- **Cloud & DevOps:** AWS (EventBridge, DynamoDB, AppSync, Cognito, SSM), Kubernetes, Docker, CI/CD (CircleCI, ArgoCD, Jenkins), Grafana, Observability & Monitoring
- **Databases & Data Stores:** PostgreSQL, MySQL, MongoDB, DynamoDB, Elasticsearch
- **Security & Compliance:** OTP & 2FA, JWT, Rate Limiting, Audit Logging, KYC, Financial Data Integrity
- **Mobile Development:** Android (Kotlin, Java), iOS (Swift), Firebase
- **Applied AI:** AI Research & Prototyping, LLM-Assisted Developer Tooling, AI for Compliance & Security

PROFESSIONAL EXPERIENCE

Cashia Limited | Senior Software Engineer (Contract)

Remote | May 2025 – Present

Fintech platform for P2P and P2M payments serving merchants and consumers.

Architecture & System Design

- Owned end-to-end architecture for the Notifications domain across Merchant Portal SDK, Consumer App, and Notifications Service, standardizing delivery workflows, rate limiting, and preference management.
- Designed resilient multi-provider SMS and email integration layers (including Advanta and Tiara) featuring fallback logic and abstractions to mitigate vendor lock-in risk.
- Built the platform's custom Webhook Registration Service from scratch with JWT verification and granular event filtering, setting security standards for third-party integrations.
- Engineered **Cashia Device Nexus**, a unified device management microservice that solved tokenization inconsistencies across iOS, Android, and backend services.

- Led architectural spikes and PoCs for device workflows and SDK delivery, eliminating cross-team dependency blockers prior to production investments.

Security, Compliance & Reliability

- Defined the standardized OTP security architecture—incorporating anti-brute-force defense, rate limiting, and mandatory invalidation—now used as company-wide authentication reference.
- Spearheaded a multi-quarter OTP & Notifications Reliability Initiative, transitioning operations from reactive bug fixes to proactive rate-limiting and Grafana-based latency observability.
- Corrected complex data integrity defects across transactional pipelines, ensuring compliant records for regulatory reporting and audits.
- Served as primary responder for Sev-1 and Sev-2 production incidents, rapidly resolving critical API and OTP delivery outages to ensure service availability.

AI Research & Leadership

- Researched and prototyped AI-assisted developer tools and compliance workflows, presenting actionable PoCs for engineering productivity evaluation.
- Mentored junior and mid-level engineers, conducted detailed code/design reviews, and established documentation standards across teams.

Zepz | Senior Backend Engineer (Developer Enablement)

Remote | July 2023 – April 2025

Global digital cross-border payments enterprise operating Sendwave and WorldRemit.

- Conceptualized and built a scalable partner integration engine utilizing API gateways and message queues for receiver validation, decreasing missed transactions by 20%.
- Migrated critical payout services from distributed MongoDB to PostgreSQL, achieving 40% faster query processing, a 60% infrastructure cost reduction, and ACID-compliant data integrity.
- Enhanced operational security in Retool dashboards by engineering user action tracking (IP/location logging) and mandatory 2FA enforcement for high-risk operations.
- Modernized legacy .NET monolithic services by refactoring core modules into scalable Spring Boot microservices.
- Decoupled cross-service dependencies by consolidating application data sources to enforce microservice data ownership.
- Researched and delivered rapid AI-driven PoCs focused on automated code analysis, compliance checks, and security tooling.
- Managed Kubernetes-backed CI/CD pipelines across CircleCI, ArgoCD, and Jenkins on AWS.

Andela | Senior Backend Engineer (Client: Safaricom PLC / DigiFarm)

Remote | May 2021 – June 2023

- Lead Backend Engineer and architect for a large-scale enterprise e-commerce platform built to handle 10,000+ transactions per second (TPS).
- Architected 9 backend microservices in Node.js/TypeScript on AWS utilizing EventBridge, shared DynamoDB, API Gateway, and SSM parameters.
- Deployed highly available Elasticsearch clusters delivering 185 ms average indexing latency and 100 ms search response times.

- Integrated AWS AppSync GraphQL APIs for real-time order tracking and customer communication features.
- Supported initial rollout across 50 enterprise clients, facilitating over $200K+ in monthly transaction volume.

Irembo (via Andela) | Full Stack Engineer (Contract)

Remote | May 2023 – July 2023

- Contributed to **IremboGov**, Rwanda's primary digital government portal, developing core platform services with Java, Spring Boot, AngularJS, and AWS microservices.

Andela | Full Stack Developer & Software Engineer (Client: Mastercard Foundation Labs)

Nairobi, Kenya | March 2018 – November 2020

- Re-architected application workflows for **DukaConnect** (mobile POS using custom computer vision on low-spec Android devices), attaining an 89% recognition success rate across user test groups.
- Reduced Android APK install footprint to 14 MB via dynamic on-demand module delivery, increasing adoption in low-bandwidth regions.
- Optimized performance and stability for **Kupaa**, a Spring Boot and Angular school management platform.
- Built **Converge**, an automated room allocation system that replaced a $1M/year vendor implementation.
- Contributed to the **Mastercard Farmers Network** platform, recognized as Best AgTech Solution at the Kenya Digital Inclusion Awards (2018).

Etinx | Senior Android Developer

Remote | June 2020 – July 2021

- Executed a full codebase modernizing rewrite in 5 weeks, transitioning architecture to Google MVP pattern and migrating legacy Java to idiomatic Kotlin.
- Refactored third-party dependencies into lightweight custom routines, cutting application APK size by 22 MB.
- Led UI/UX overhauls driven by user telemetry, increasing Google Play Store app rating by +2.0 stars over 4 months.

JosLabs | Solutions Architect

Nairobi, Kenya | February 2017 – July 2019

- Led technical requirements gathering, system design, and deployment of Android applications and web platforms for transport, trade, and surveillance clients.

EDUCATION & CERTIFICATIONS

- **Bachelor of Science in Computer Science & Information Systems Security**
  Kibabii University | Graduated December 2016
- **Presidential Digital Talent Programme (PDTP)**
  ICT Authority, Kenya | 2018 Specialization: Mobile Application Development & Artificial Intelligence

ADDITIONAL INFORMATION

- **Interests:** Engineering Mentorship, Technical Writing, Hiking, Culinary Arts, DIY Hardware/IoT